
2022 RI H2 Chemistry Prelim Paper 2 – Suggested Solutions

1(a)

$$\text{Amount of iodine reacted} = \frac{65}{22700} \times \frac{1}{3} \times \frac{3}{2} = 1.432 \times 10^{-3} \text{ mol}$$

$$\text{Mass of iodine} = 1.432 \times 10^{-3} \times 126.9 \times 2 = \underline{0.363 \text{ g}}$$

1(b)(i)

The boiling points increase from HCl to HBr to HI as the size of electron cloud for polarisation increases, resulting in stronger instantaneous dipole-induced dipole (id-id) interactions, and hence requiring more energy to overcome.

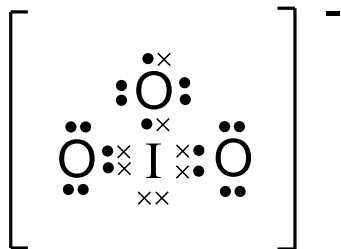
The boiling point decreases from HF to HCl (or HF has the highest boiling point) due to the presence of stronger hydrogen bonding between HF molecules which requires more energy to overcome.

1(b)(ii)

Down Group 17, the valence orbital of the halogen atom becomes increasingly diffuse and the orbital overlap between the halogen and hydrogen atoms becomes less effective / electronegativity difference between the halogen and hydrogen decreases, resulting in a decrease in bond polarity.

Hence, the H–X bond strength decreases from HCl to HBr to HI and the thermal stability decreases from HCl to HBr to HI.

1(c)(i)



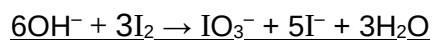
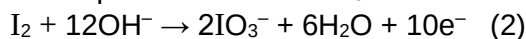
1(c)(ii)

In IO_3^- , there are 3 bond pairs and 1 lone pair of electrons around the central I atom. To minimise electronic repulsion between the bond pairs and lone pair electrons, the shape about the I atom is trigonal pyramidal.

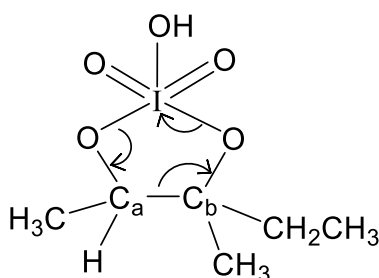
As the lone pair-bond pair repulsion is greater than the bond pair-bond pair repulsion, the bond angle is 107° .

1(c)(iii) Method 1: Oxidation number methodOxidation number of I in I_2 = 0Oxidation number of I in IO_3^- = +5Oxidation number of I in I^- = -11 mol of I atom loses 5 mol of e^- to form 1 mol of IO_3^- .1 mol of I atom gains 1 mol of e^- to form 1 mol of I^- .

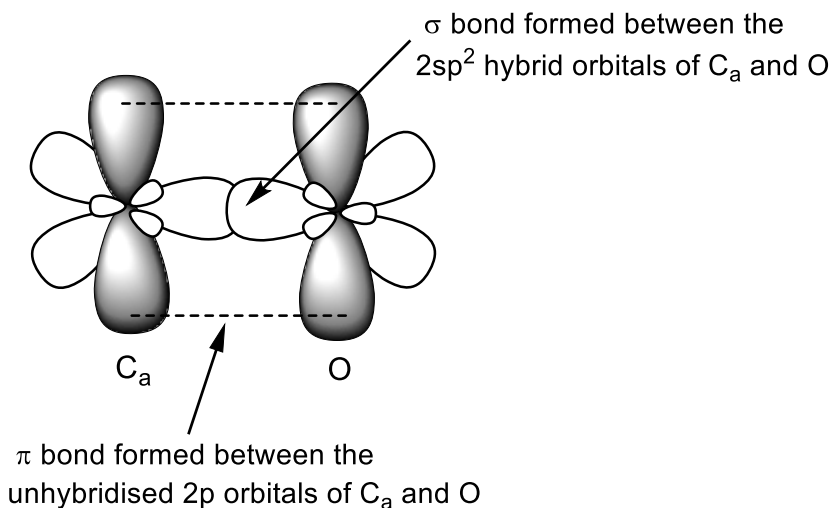
Since

no. of e^- lost = no. of e^- gained,mole ratio of $IO_3^- : I^-$ = 1 : 5**Method 2: Half-equation method**Half equation from I_2 to I^- :Half equation from I_2 to IO_3^- :

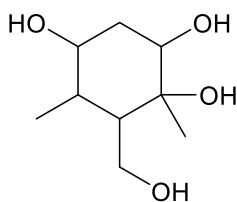
Balancing equation (1) and (2):

**1(d)(i)**intermediate **B****1(d)(ii)** The C_a -H bond in ethanal is stronger than that in glycol **A**.

The $2sp^2$ orbital of C_a in ethanal has higher s character and is less diffuse than the $2sp^3$ orbital of C_a in glycol **A**. Hence, the $2sp^2$ orbital of C_a in ethanal has a more effective overlap with the 1s orbital of H leading to a stronger bond.

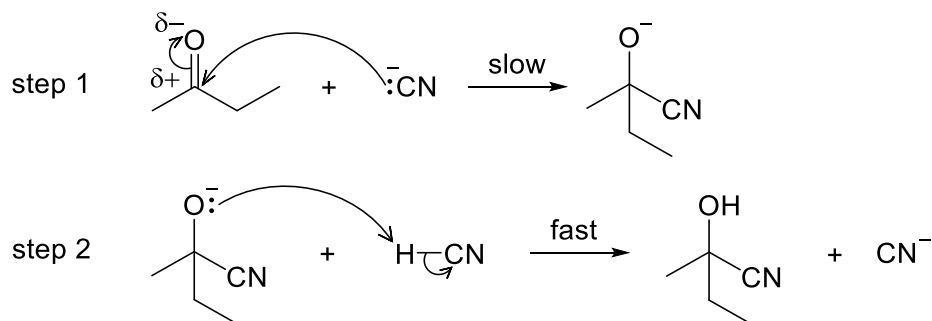
1(d)(iii)

1(d)(iv)



1(d)(v)

Nucleophilic addition

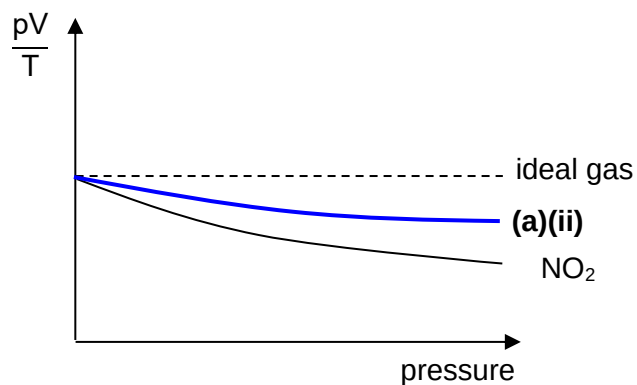


2(a)(i)

Any three of the following:

- An ideal gas consists of particles of negligible volume.
- The gas particles exert negligible attractive forces on one another.
- The gas particles are in constant random motion.
- Collisions between the gas particles are perfectly elastic.
- The average kinetic energy of particles in a gas is constant at constant temperature. OR The average kinetic energy is proportional to the absolute temperature.

2(a)(ii)



At higher temperature of T_2 K, the NO_2 gas molecules have higher kinetic energy and are better able to overcome the intermolecular attractive forces. Hence it would exhibit less deviation from an ideal gas as compared to at a lower temperature of T_1 K.

2(b)(i) At $t = 200 \text{ s}$, $[\text{NOCl}] = 0.032 \text{ mol dm}^{-3}$

	$2\text{NO}_2(\text{g})$	+	$\text{HCl}(\text{g})$	$\longrightarrow$	$\text{NOCl}(\text{g})$	+	$\text{HNO}_3(\text{g})$
initial conc / mol dm^{-3}	1.00		0.05		0		0
change in conc / mol dm^{-3}	-0.064		-0.032		+0.032		+0.032
conc at 200 s / mol dm^{-3}	0.936		0.018		0.032		0.032

$$\text{mole fraction of NOCl} = \frac{0.032}{0.936 + 0.018 + 0.032 + 0.032} = \underline{0.0314}$$

2(b)(ii) $pV = nRT$

$$p_{\text{NOCl}}V = n_{\text{NOCl}}RT$$

$$\begin{aligned} p_{\text{NOCl}} &= \frac{n_{\text{NOCl}}}{V} RT \\ &= \frac{0.032}{10^{-3}} (8.31)(500) \\ &= 1.3296 \times 10^5 \text{ Pa} \end{aligned}$$

$$p_{\text{NOCl}} = x_{\text{NOCl}} \times p_{\text{total}}$$

$$\begin{aligned} p_{\text{total}} &= \frac{1.3296 \times 10^5}{0.0314} \\ &= \underline{4.23 \times 10^6 \text{ Pa}} \end{aligned}$$

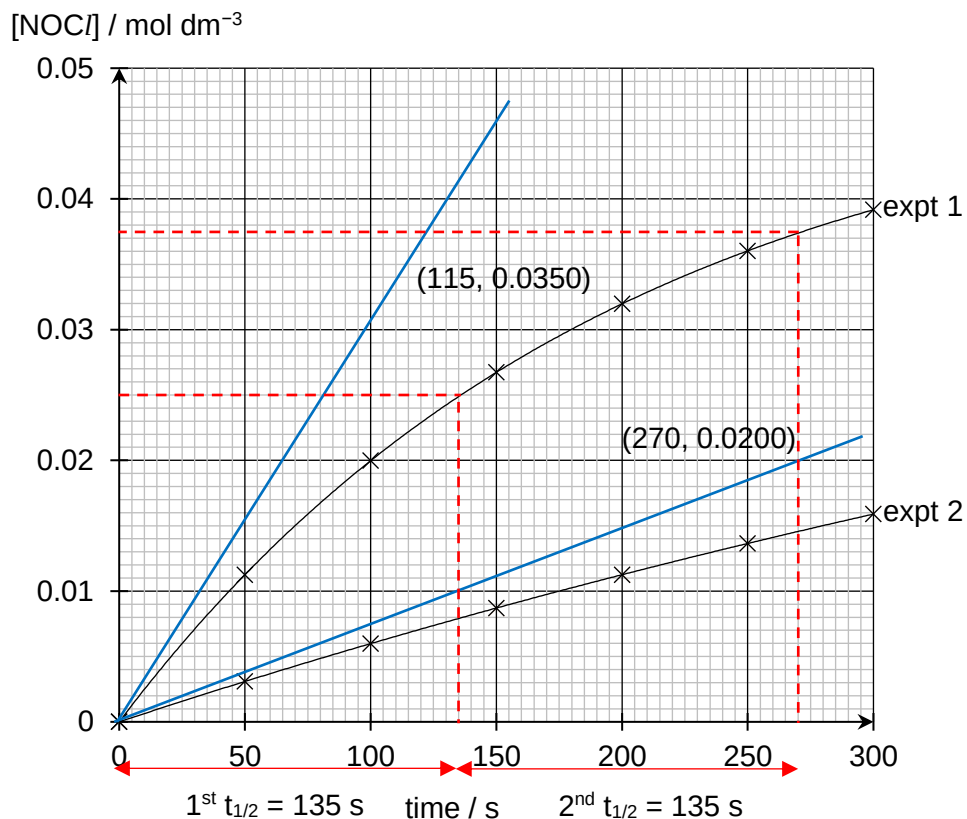
Alternative method:

$$pV = nRT$$

$$p_{\text{total}}V = n_{\text{total}}RT$$

$$\begin{aligned} p_{\text{total}} &= \frac{n_{\text{total}}}{V} RT \\ &= \frac{0.936 + 0.018 + 0.032 + 0.032}{10^{-3}} (8.31)(500) \\ &= \underline{4.23 \times 10^6 \text{ Pa}} \end{aligned}$$

2(b)(iii)



$$\text{initial rate of reaction for experiment 1} = \frac{0.0350 - 0}{115 - 0} = 3.04 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$$

$$\text{initial rate of reaction for experiment 2} = \frac{0.0200 - 0}{270 - 0} = 7.41 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$$

$$\frac{\text{initial rate of expt 1}}{\text{initial rate of expt 2}} = \frac{3.04 \times 10^{-4}}{7.41 \times 10^{-5}} = 4.10 \approx 4 \text{ (to nearest integer)}$$

Comparing experiments 1 and 2, when $[\text{NO}_2] \times 2$, rate $\times 4$.

$\Rightarrow \text{rate} \propto [\text{NO}_2]^2$,

$\Rightarrow$ reaction is second order with respect to NO_2 .

2(b)(iv) Final $[\text{NOCl}] = \underline{0.0500 \text{ mol dm}^{-3}}$

2(b)(v) For experiment 1, final $[\text{NOCl}] = 0.05 \text{ mol dm}^{-3}$.

1st $t_{1/2} \Rightarrow$ time taken for $[\text{HCl}]$ to decrease from 0.05 mol dm^{-3} to $0.025 \text{ mol dm}^{-3}$
 $\Rightarrow$ time taken for $[\text{NOCl}]$ to increase from 0 mol dm^{-3} to $0.025 \text{ mol dm}^{-3}$
 $\Rightarrow 135 \text{ s}$

2nd $t_{1/2} \Rightarrow$ time taken for $[\text{HCl}]$ to decrease from $0.025 \text{ mol dm}^{-3}$ to $0.0125 \text{ mol dm}^{-3}$
 $\Rightarrow$ time taken for $[\text{NOCl}]$ to increase from $0.025 \text{ mol dm}^{-3}$ to $0.0375 \text{ mol dm}^{-3}$
 $\Rightarrow 135 \text{ s}$

Since $t_{1/2}$ is constant, reaction is first order with respect to HCl .

2(b)(vi) Since NO_2 is in large excess, rate = $k'[\text{HCl}]$ where $k' = k[\text{NO}_2]^2$.

$$\text{Hence } t_{1/2} = \frac{\ln 2}{k'} = \frac{\ln 2}{k[\text{NO}_2]^2}.$$

OR

Since NO_2 is in large excess, the overall reaction is pseudo first-order with respect to HCl . Hence $t_{1/2}$ is independent of $[\text{HCl}]$.

As initial $[\text{NO}_2]$ of experiment 3 is the same as that of experiment 2, $t_{1/2}$ would remain the same at 540 s even though $[\text{HCl}]$ is different.

2(c)(i) $\text{N}_2\text{O}_4 + \text{CH}_2\text{CH}_2 \longrightarrow \text{O}_2\text{NCH}_2\text{CH}_2\text{NO}_2$

2(c)(ii) Addition reaction

2(c)(iii) The N_2O_4 molecules diffuse to the surface of the Pt catalyst and they become adsorbed onto the active sites of the Pt surface.

The adsorption weakens the covalent bonds in the N_2O_4 molecules and lowers the activation energy.

Once the NO_2 molecules are formed, they desorb from the surface and diffuse away from the Pt catalyst, freeing up the active sites for adsorbing other N_2O_4 molecules.

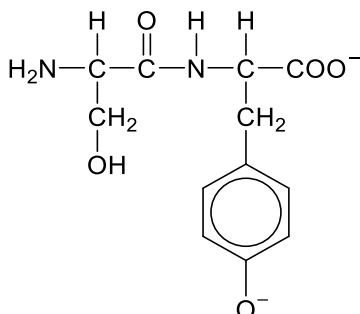
2(c)(iv) $K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$

$$[\text{NO}_2] = \sqrt{K_c [\text{N}_2\text{O}_4]} \text{ OR } K_c^{1/2} [\text{N}_2\text{O}_4]^{1/2}$$

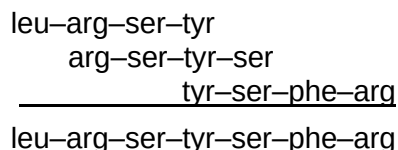
2(c)(v) rate = $k_2[\text{NO}_2][\text{CH}_2\text{CH}_2]$
 rate = $k'[\text{N}_2\text{O}_4]^{1/2}[\text{CH}_2\text{CH}_2]$ where $k' = k_2 K_c^{1/2}$

3(a)(i) Upon heating with NaOH , the amide in the side chain of glutamine was hydrolysed to form $-\text{CH}_2\text{CH}_2\text{COO}^-$. Upon acidification, the $-\text{CH}_2\text{CH}_2\text{COO}^-$ undergoes protonation to form $-\text{CH}_2\text{CH}_2\text{COOH}$ which is the side chain of glutamic acid.

3(a)(ii)



3(a)(iii)



3(b)(i)

(CH₃)₃N is more basic as it has 2 more electron-donating alkyl groups, which increase the electron density on the N atom, making the lone pair of electrons on N more available for coordination to an electron deficient species / a proton.

3(b)(ii)

ion-dipole interactions

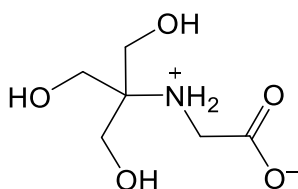
3(b)(iii)

Due to its smaller ionic radius/size, CH₃NH₃⁺ has a higher charge density compared to (CH₃)₃NH⁺. Hence, CH₃NH₃⁺ forms stronger ion-dipole interactions with water.

3(b)(iv)

The stronger interaction between CH₃NH₃⁺ and water leads to greater stability and causes the position of equilibrium of equation 3.2 to lie more to the right / greater extent of ionisation of CH₃NH₂ as compared to (CH₃)₃N. Hence, CH₃NH₂ is a stronger base with a lower pK_b.

3(c)(i)

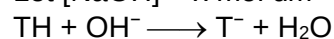


TH exists as a zwitterion with oppositely charged ends. It has a giant ionic lattice structure consisting of strong ionic bonds between the oppositely charged ends of neighbouring zwitterions. The strong ionic bonds require a large amount of heat energy to overcome. Hence, TH has a high melting point and exists as a solid at room temperature.

3(c)(ii)

$$\text{pH} = \text{p}K_a + \lg \frac{[\text{T}^-]}{[\text{TH}]}$$

$$8.78 = 8.15 + \lg \frac{[\text{T}^-]}{[\text{TH}]}$$

Let $[\text{NaOH}] = x \text{ mol dm}^{-3}$ 

$$\frac{[\text{T}^-]}{[\text{TH}]} = 10^{0.63} = \frac{x}{0.1-x}$$

$$x = 0.08101$$

$$[\text{NaOH}] = \underline{0.0810 \text{ mol dm}^{-3}}$$

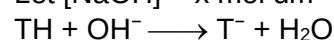
OR

$$\text{pOH} = \text{p}K_b + \lg \frac{[\text{TH}]}{[\text{T}^-]}$$

$$\text{p}K_b = 14 - 8.15 = 5.85$$

$$\text{pOH} = 14 - 8.78 = 5.22$$

$$5.22 = 5.85 + \lg \frac{[\text{TH}]}{[\text{T}^-]}$$

Let $[\text{NaOH}] = x \text{ mol dm}^{-3}$ 

$$\frac{[\text{TH}]}{[\text{T}^-]} = 10^{-0.63} = \frac{0.1-x}{x}$$

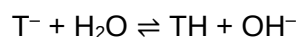
$$x = 0.08101$$

$$[\text{NaOH}] = \underline{0.0810 \text{ mol dm}^{-3}}$$

3(c)(iii)

With lower concentrations of T^- and TH , the buffering capacity will decrease as there is less amount of T^- and TH to react with added H^+ or OH^- respectively.

3(d)(i)



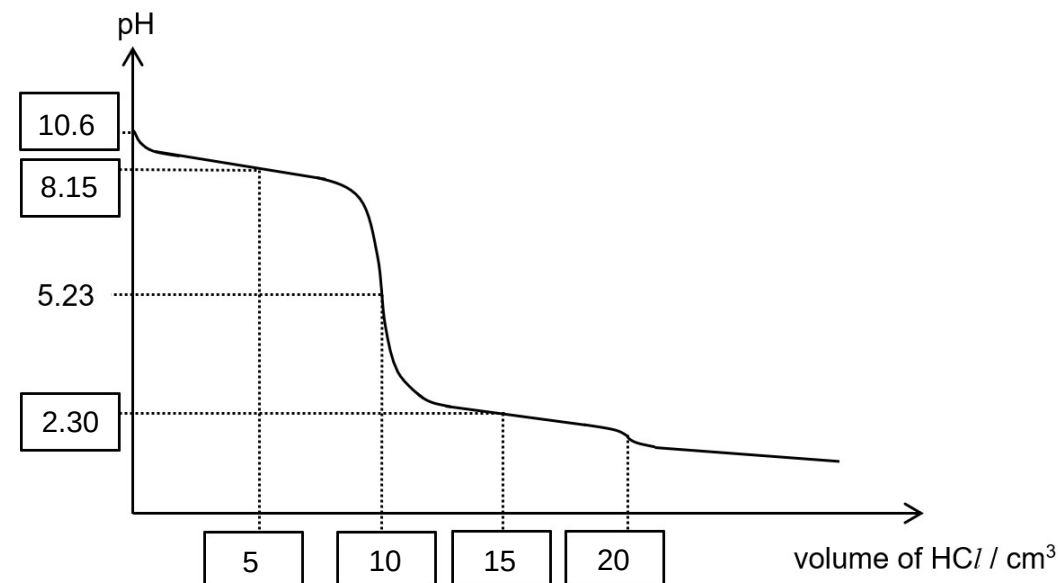
$$\text{p}K_{b2} = 14 - 8.15 = 5.85$$

$$[\text{OH}^-] = \sqrt{10^{-5.85} \times 0.1} = 3.758 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{pOH} = -\lg(3.758 \times 10^{-4}) = 3.425$$

$$\text{pH} = 14 - 3.425 = \underline{10.6}$$

3(d)(ii)



3(d)(iii)

The pH range of methyl red coincides with the region of rapid pH change in the first equivalence point. At the first end-point, the colour change is from yellow to orange.

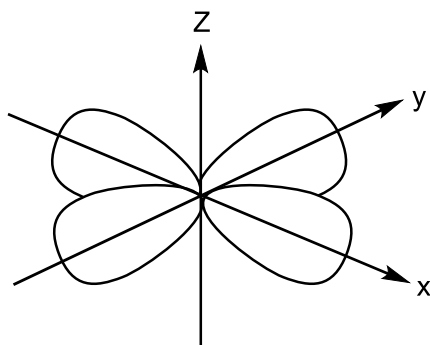
3(d)(iv)

There is no region of rapid pH change at the second equivalence point.

4(a)(i) Since the proportion of **G** to **F** increases with temperature, the forward reaction is favoured by an increase in temperature. As increase in temperature favours the endothermic reaction which absorbs heat, the forward reaction is endothermic and hence y is a positive value.

4(a)(ii) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$

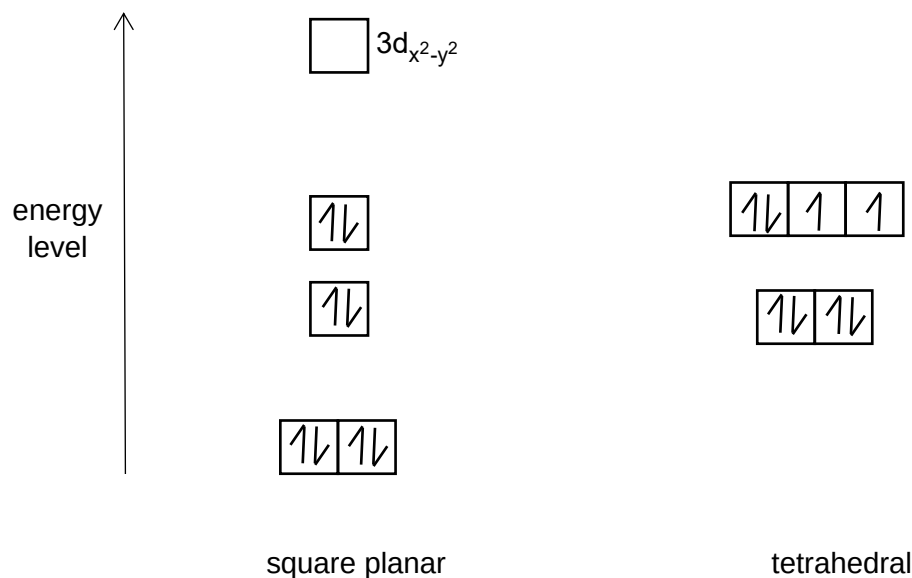
4(a)(iii)



In **F** (a square planar complex), the four ligands approach Ni^{2+} (the central metal ion) along the x and y axes.

Since $3d_{x^2-y^2}$ orbital has its lobes along the x and y axes, any electron in the $3d_{x^2-y^2}$ orbital would be repelled most greatly by the lone pair of electrons in the ligands, causing $3d_{x^2-y^2}$ orbital to have the highest energy.

4(a)(iv)

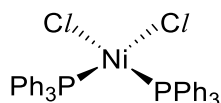


- 4(a)(v)** In **F** and **G**, the 3d orbitals are split differently, causing the energy gap between the higher and lower energy levels to be different.

The energy required to promote an electron from the lower energy to the higher energy d orbital corresponds to different wavelengths of light.

The different colours observed of **F** and **G** are the complement of the colour absorbed.

- 4(a)(vi)**



H has two large PPh₃ groups which are close to each other. This results in great repulsion / steric strain between the two groups, causing **H** to be less stable than **F**. Hence, **G** is converted to the more stable **F** rather than **H**.

- 4(b)(i)** X⁻ must have a large non-polar/ hydrophobic portion to enable CuX₂ to interact well with oil rather than with water.

- 4(b)(ii)** Concentration of CuX₂(oil) at equilibrium = 0.045 mol dm⁻³

- 4(b)(iii)**

$$K_c = \frac{[\text{CuX}_2(\text{oil})][\text{H}^+(\text{aq})]^2}{[\text{Cu}^{2+}(\text{aq})][\text{HX}(\text{oil})]^2}$$

$$2.1 \times 10^{-2} = \frac{(0.045)(10^{-3})^2}{[\text{Cu}^{2+}(\text{aq})](0.20)^2}$$

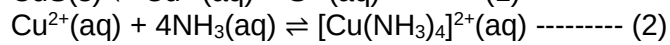
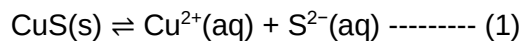
$$[\text{Cu}^{2+}(\text{aq})] \text{ at equilibrium} = 5.36 \times 10^{-5} \text{ mol dm}^{-3}$$

$$\begin{aligned} \text{Initial amount of Cu}^{2+} \text{ in } 2.0 \text{ dm}^3 &= \text{eqm amt of Cu}^{2+}(\text{aq}) + \text{eqm amt of CuX}_2(\text{oil}) \\ &= (2 \times 5.36 \times 10^{-5}) + (1 \times 0.045) \text{ mol} \\ &= 0.0451 \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Percentage of copper extracted} &= \frac{\text{eqm amount in oil}}{\text{initial amount}} \\ &= \frac{0.045}{0.0451} \times 100\% = \underline{99.8\%} \end{aligned}$$

- 4(b)(iv)** Cu²⁺ can be released from the CuX₂ molecules in the oil solution by adding H₂SO₄ to increase [H⁺(aq)]. This shifts the position of equilibrium in equation 4.2 to the left, converting CuX₂(oil) to Cu²⁺(aq).

4(c)



Cu^{2+} reacts with NH_3 in the metal polish to form a soluble complex $[\text{Cu}(\text{NH}_3)_4]^{2+}$, thus decreasing $[\text{Cu}^{2+}]$.

This causes the position of equilibrium in (1) to shift to the right so that the tarnish, CuS, dissolves and can be removed.

OR

If there is sufficient NH_3 , the ionic product, $[\text{Cu}^{2+}][\text{S}^{2-}]$, will be less than the K_{sp} and hence all the CuS dissolves.